

Abnormal Urine

College of Medicine - HMU
1st year Medical Students

Prepared by
Shelan I. Saber
MSc. in Biochemistry

Test for some of the abnormal (Pathological) constituents of urine.

Substances which are not present in easily detectable quantities in urine of healthy person, but occur in urine under certain disease conditions, are called abnormal or pathological constituent of urine.

The detection of these substances in urine is the matter of great clinical importance.

The symptoms for diagnosis and treatment can be judged more accurately by the quantitative estimation of those abnormal constituent.

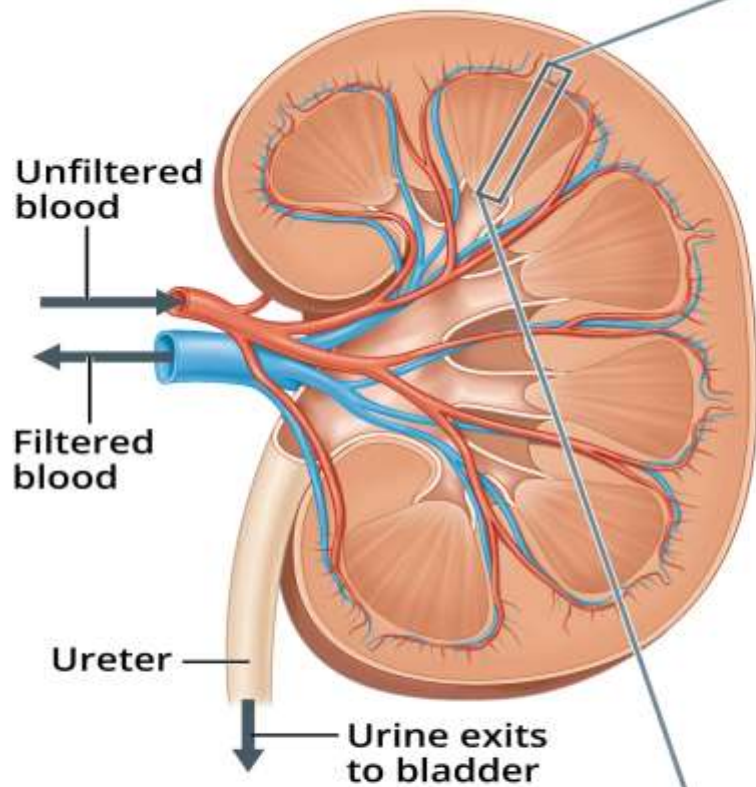
1- Protein:-

Normally non detectable amounts of protein are excreted in urine, presence of detectable amount of protein in urine is known as proteinuria or albuminuria of which we have the following types:-

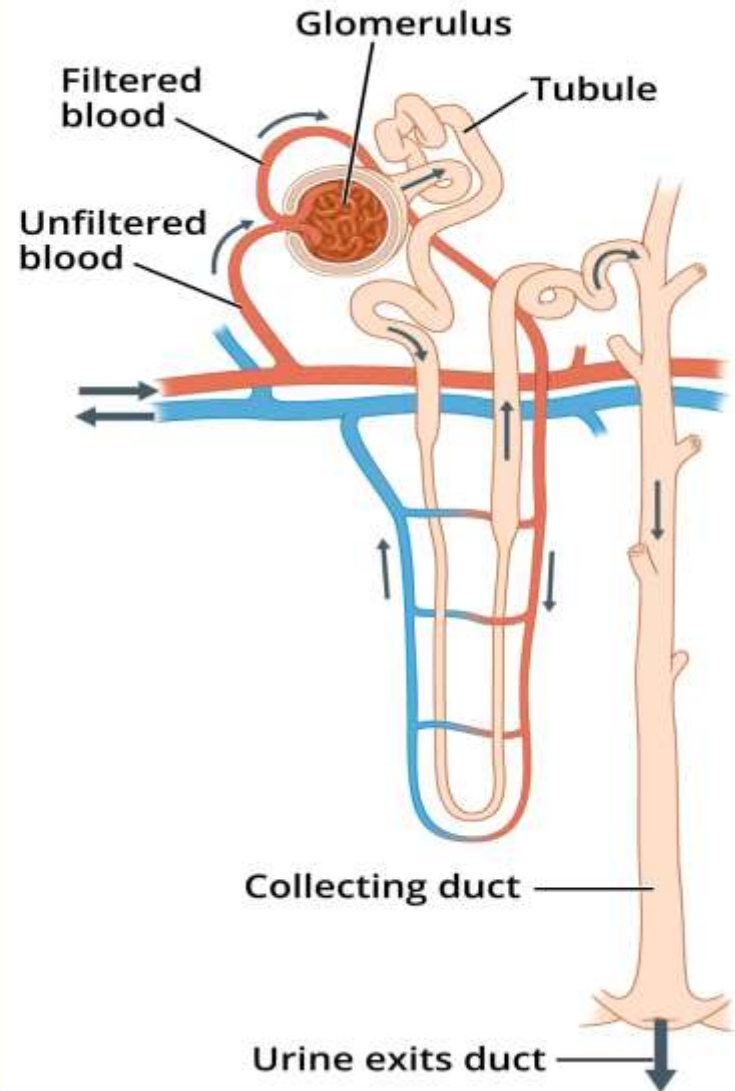
Pathological proteinuria in renal disease, this is due to serum proteins passing through glomeruli in nephritis or glomerulonephritis.

Physiological proteinuria this caused by excessive muscular exercise, pregnancy and after high protein meal.

Kidney

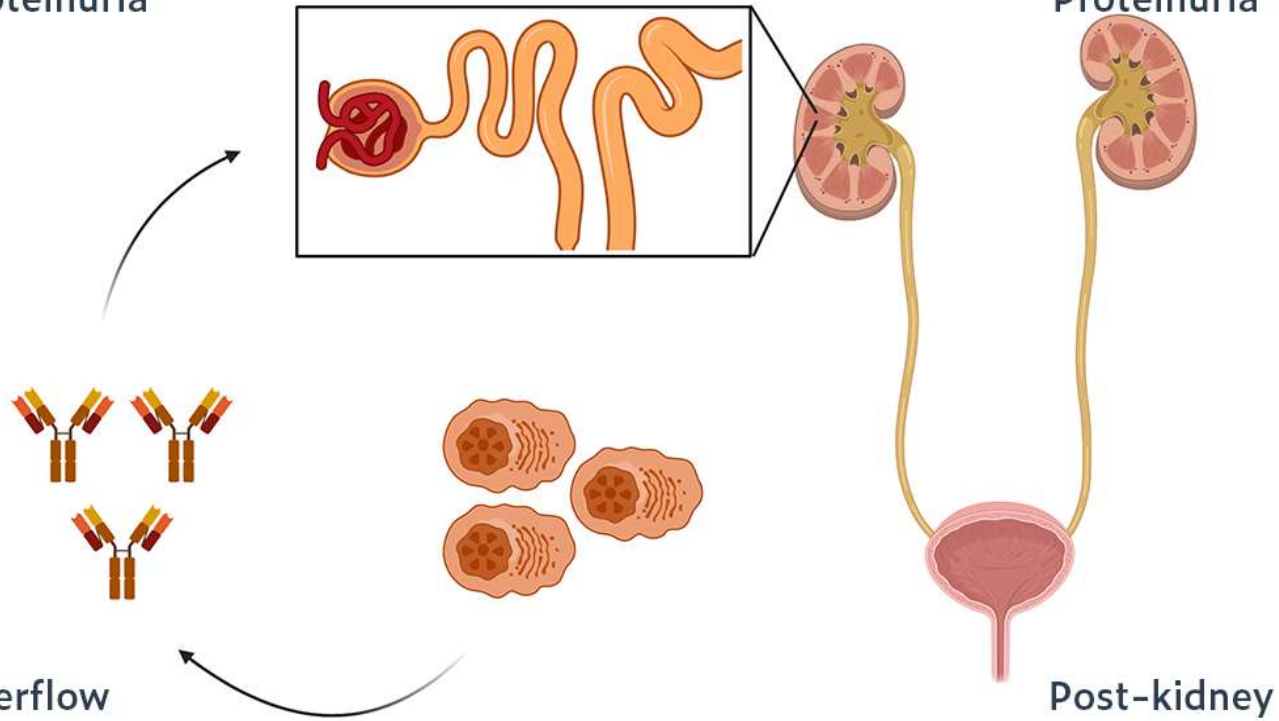


Nephron



Glomerular Proteinuria

Tubular Proteinuria



Overflow Proteinuria

Post-kidney Proteinuria

Created by Dr. Ayman Al Jurdi using BioRender.com

GlomCon^{edu}

2- Sugar / Glucose:-

Glucose is present normally to the extent of 0.01 – 1 gm per a day in urine, and this quantity is not sufficient enough to give a positive reaction in the qualitative tests.

Presence of large amount of glucose in urine is called glycosuria, finding of glycosuria indicates that the person is hyperglycemic or has a lowered renal threshold for glucose. The renal threshold for glucose is approximately 160 to 190 mg/100ml of blood glucose; and does not appear in the urine until the blood glucose rises above this level.

3- Ketone bodies:-

The presence of ketone bodies in the urine is known as ketonuria. Ketonuria may accompany any disorders, the supply of glycogen in the liver is rapidly depleted and the energy is derived from stored fats. Increased catabolism of fats in the absence of sufficient carbohydrates results in excessive formation of ketone bodies.

Uncontrolled
D.mellitus

Ketone bodies

Low insulin
leads to

- Increased lipolysis
- Decreased reesterification

increased Ketone
bodies

Increased plasma
free fatty acids

Reesterification

Triglycerides

Stored in liver

Low density lipoprotein

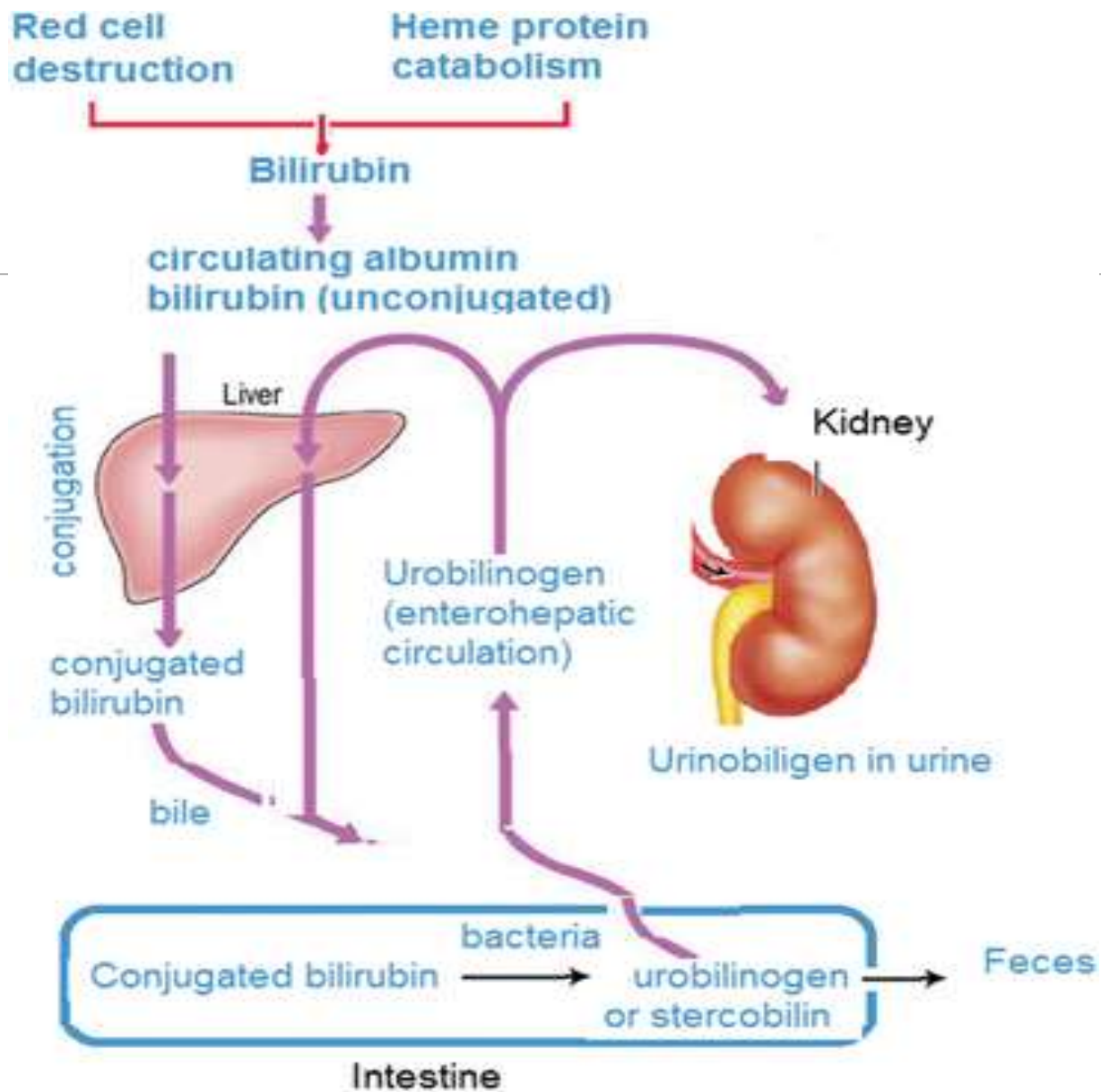
1. Acetone
2. Acetoacetate
3. beta-hydroxybutaric acid

4- Bile pigment (Bilirubin)

The presence of bile pigment (bilirubin) in urine is usually due to an obstructive jaundice of the bile duct by calculi and also they may be found in hepatitis jaundice.

Hemolytic jaundice (Un-conjugated) dose not cause bile pigment to enter the urine. This point is used as a general differential diagnosis between obstructive and hemolytic jaundice.

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5- Tests for blood in urine (Hematuria) :-

Red blood cells are seen in acute glomerular nephritis, infective endocarditis, destructive lesions of urinary tract like tumors, calculi, tuberculosis, trauma and infection of kidney.